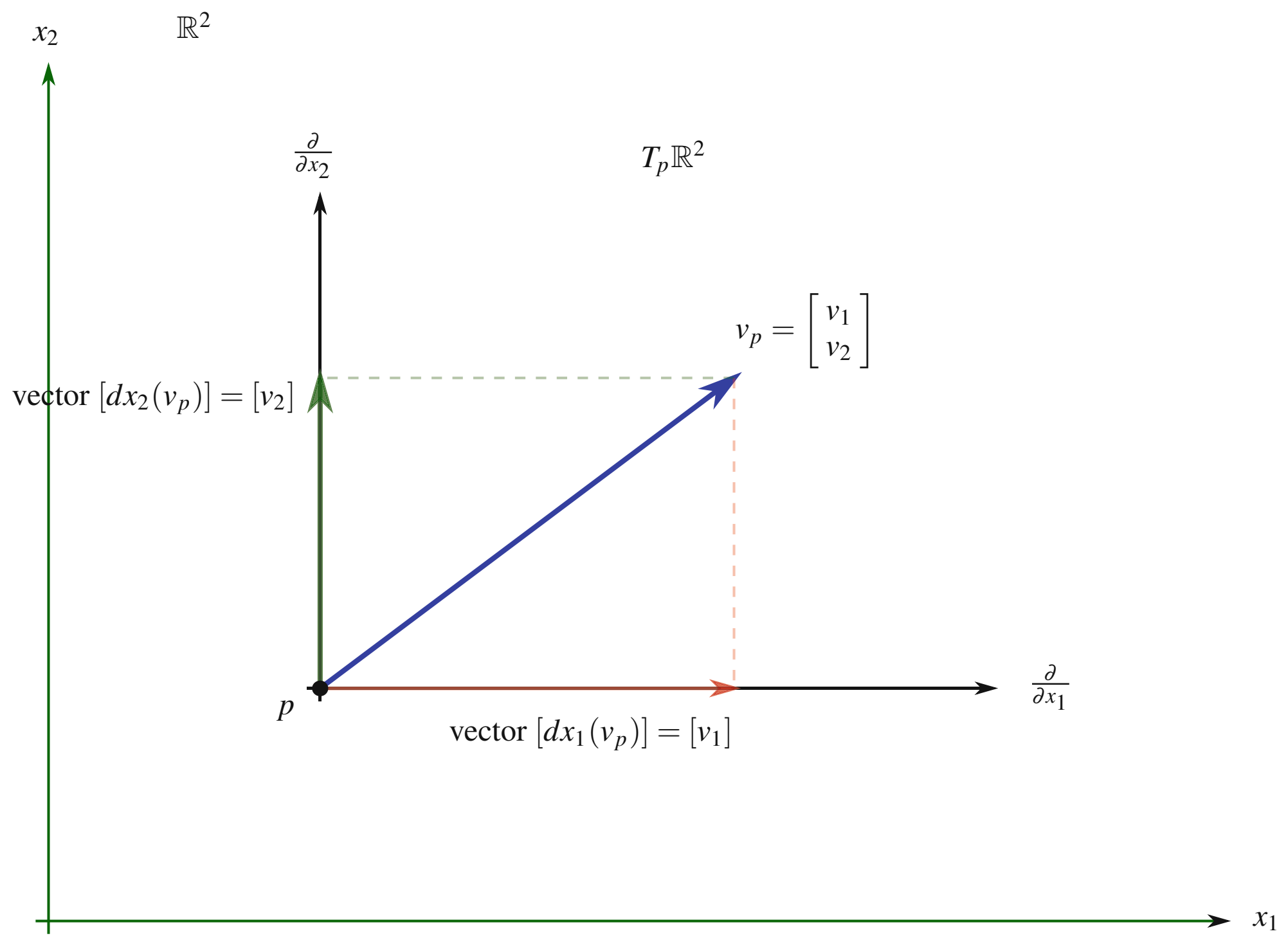

$x_2$
$\mathbb{R}^2$
$\frac{\partial}{\partial x_2}$
$T_p\mathbb{R}^2$
$v_p = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$
vector $[dx_2(v_p)] = [v_2]$
$p$
$\frac{\partial}{\partial x_1}$
vector $[dx_1(v_p)] = [v_1]$
$x_1$